

Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/MICROBIOLOGY/BIOCHEMISTRY)

SEMESTER I EXAMINATION

BT/ MI/ BC 703: BIOINSTRUMENTATION AND STATISTICS

Date: 05/05/2010

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A

Total marks: 20

Q1. Choose the correct option and put $\checkmark$ mark in front of it:

- 1) The ability of Microscope to distinguish two objects into two separate objects is called
a) resolving power b) Wavelength c) N.A. d) None
- 2) Living, unstained cells and organisms can be observed best using
A) fluorescent microscopy B) TEM C) Phase contrast microscopy D) SEM
- 3) Which of the following is NOT equivalent to 10 micrometers.
A) 0.0001 cm B) 0.01 mm C) 10,000 nm D) 100,000 Angstroms
- 4) The phenomenon whereby a molecule, after absorbing radiations of shorter wavelength, emits radiation of a longer wavelength is known as:
(a) Luminescence (b) Transmission (c) Fluorescence (d) Emission
- 5) The region of an infra-red spectrum where many absorptions take place is known as the...
a) thumbprint region b) handprint region
c) footprint region d) fingerprint region
- 6) Beer's Law is related to
a) Concentration b) Path length c) Both a) and b)
d) None of the above
- 7) The energy E, of a photon can be related to its wavelength and frequency with the help of plank's constant h, as
a) $E = h c / \lambda$ b) $E = c / h \lambda$ c) $E = h \lambda / C$ d) $E = \lambda / h c$
- 8) The chemical shift in NMR is because of
a) Shielding and deshielding b) Solvent effects
c) Hydrogen bonding d) All of the above

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- 9) The first dimension of separation for two-dimensional electrophoresis is based on
a) molecular mass b) isoelectric point c) folding d) Size
- 10) For an enzyme to be separated by affinity chromatography, following should NOT be used as a ligand...
(a) Effectors (b) Co-factors (c) Receptor antagonist (d) Substrate
- 11) In GC, halogenated molecules can be analyzed using the following detector,
(a) Flame ionization detector (b) Electron capture detector
(c) UV-Vis Spectrophotometer (d) Thermionic emission detector
- 12) The principle of separation in GLC is
a) adsorption b) Partition c) Ion Exchange d) Gel permeation
- 13) Who first developed the DNA sequencing approach using dideoxynucleoside triphosphates in DNA synthesis?
A) Pauling B) Watson C) Sanger D) Edman's
- 14) Small solid supports onto which are spotted hundreds of thousands of tiny drops of DNA that can be used to screen gene expression.
a) DNA microarrays b) cloning library c) Southern blot d) western blot
- 15) Sequenator is an automatic machine to determine amino acid sequence in a polypeptide chain. The reagent used in sequenator is
a) sanger's reagent b) CNBr c) Trypsin d) Edman's Reagent
- 16) PCR is used for
a) digesting proteins b) digesting DNA
c) amplifying DNA d) amplifying proteins
- 17) The mode of the observations in data set is the observation that occurs maximum number of times (True/False)
- 18) The mean of Binomial distribution with parameters n and p is _____
- 19) The median of a data set is the observation such that 25% of the observations are less than or equal to it (True/False)
- 20) Potato cyst-nematode (PCN) is a serious pest of potatoes. It can be controlled by chemical nematicides, by deep cultivation, or by planting varieties resistant to PCN. In a field experiment to study the effects of these methods of control, the data are obtained on following variables:
Variety: Resistant (R) v Susceptible (S)
Cultivation: Deep (D) v Normal (-)
Nematicide: None (-) v Nematicide (N)
Mean yields of potato tubers (tonnes/hectare)
- The scale of these variables are _____ and _____ (nominal, ordinal, ratio,)

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2. Write answers for Section I and Section II of Part B in separate answer sheets.

Part B

Total Marks: 50

Section I

- Q1 A Write short note on: (Any One)
- a) Isolation and purification of plasmid DNA [5]
 - b) PCR [5]
- B Briefly discuss DNA sequencing Methods [5]

- Q2 A The data describe an experiment conducted to investigate the amount of drug present in the liver of a rat. Ten rats were randomly selected, weighed, and placed under a light anesthetic and given an oral dose of the drug. Because the it was thought that large livers would absorb more of a given dose than a small liver, the actual dose given was approximately determined as 40mg of the drug per kilogram of body weight. After a fixed length of time, each rat was sacrificed and the liver weighed, and the percent dose in the liver was determined. [5]

Table: Drug In Liver(The proportion of the dose in the liver)

0.42 0.25 0.56 0.23 0.23 0.32 0.37 0.41 0.33 0.38

Calculate,

Mean dose in liver

Standard deviation of dose in liver.

OR

- A The weights in grams of 60 seedling of a plant when taken from a greenhouse for planting out were as below. [5]

21 18 33 40 20 19 38 47 26 32
15 38 31 36 34 25 28 29 22 42
45 24 27 40 19 16 45 15 42 28
15 31 33 28 42 37 40 29 24 25
29 29 40 35 32 36 34 41 27 31
18 31 19 33 25 26 20 36 27 37

Organize the data by means of frequency distribution using class width 5.

- B A batch of 5000 tablets has been produced and the concentration of active agent assayed in all tablets. The mean ($\pm\sigma$) of the batch is 200 ± 10 mg and the distribution of concentrations of active agent is normally distributed. Calculate the proportion of the batch that contains 180 mg or less of active agent. [5]

Section II

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|----|---|--|-----|
| Q3 | A | Briefly discuss phase contrast microscopy | [5] |
| | | OR | |
| | A | Draw ray diagram of SEM | [5] |
| | B | Discriminate between ELISA and RIA | [3] |
| | C | Write principle of Fluorescent microscopy | [2] |
| Q4 | A | Discuss following components of Visible spectroscopy | [5] |
| | | a) Photomultiplier | |
| | | b) Prism | |
| | | OR | |
| | A | Explain Chemical Shift | [5] |
| | B | Discuss basic principle of Mass Spectroscopy. Enlist its application | [5] |
| Q5 | A | Write short notes on: (Any two) | [5] |
| | | a) Ion exchange chromatography | [5] |
| | | b) Measurement of Viscosity | [5] |
| | | c) PAGE | [5] |